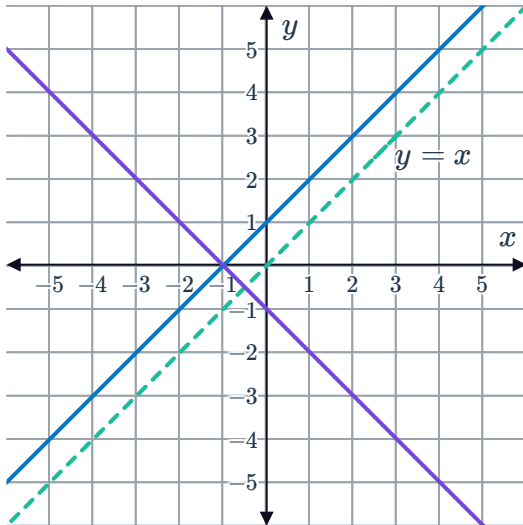


Inverse relations and functions

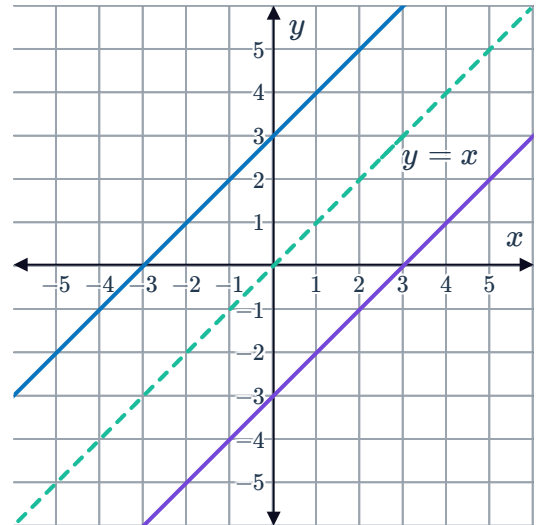


- 1 The graphs of a function f and its inverse f^{-1} are symmetric about which line?
- 2 In each graph below two functions and the line $y = x$ are drawn. For each graph, state whether the functions are inverse functions of each other.

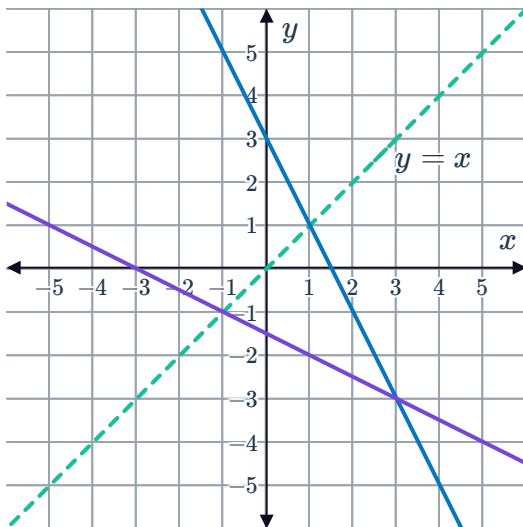
a



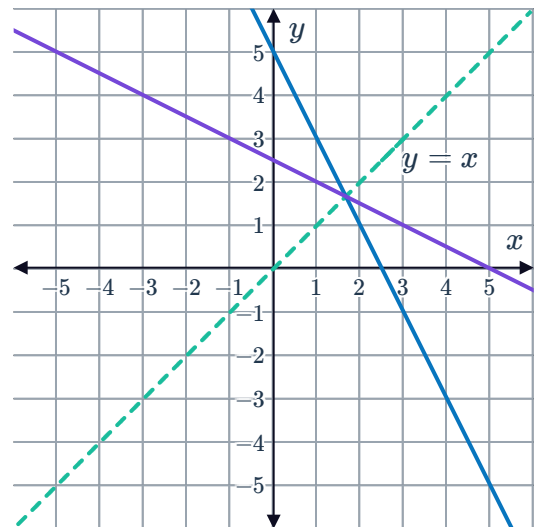
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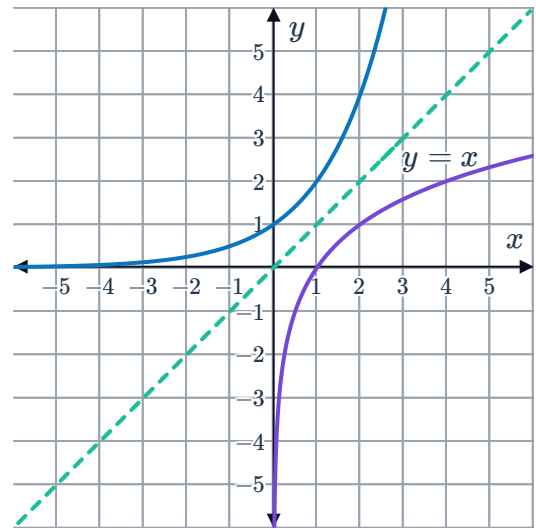
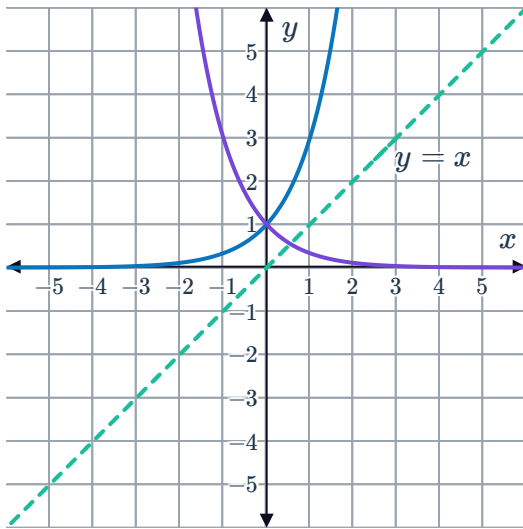
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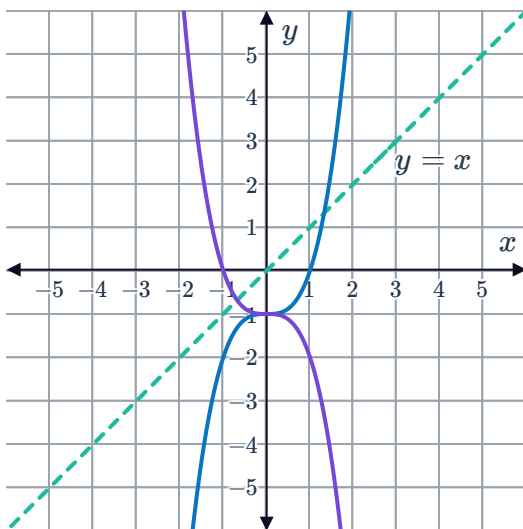
d



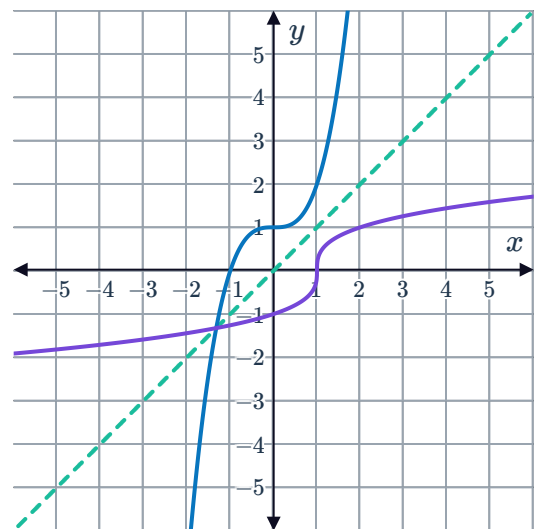
e



g

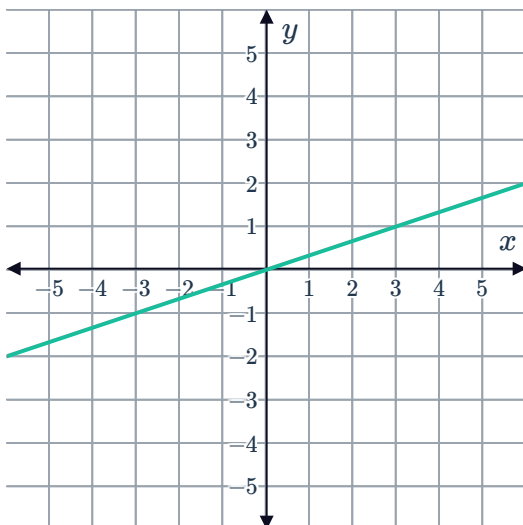


h

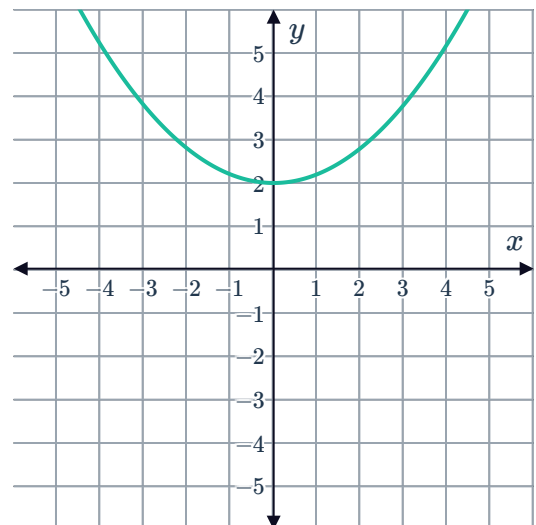


3 State whether the following functions have an inverse function:

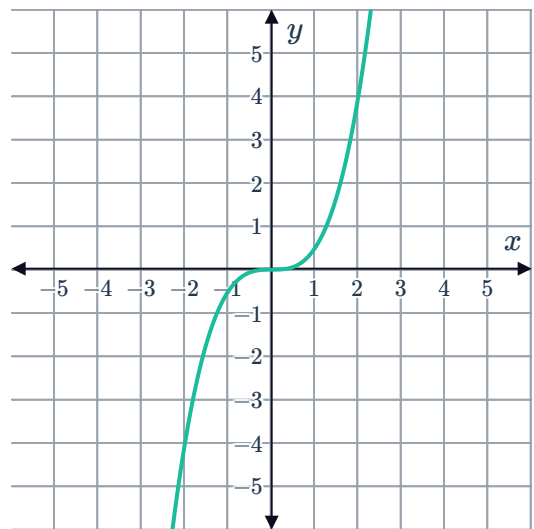
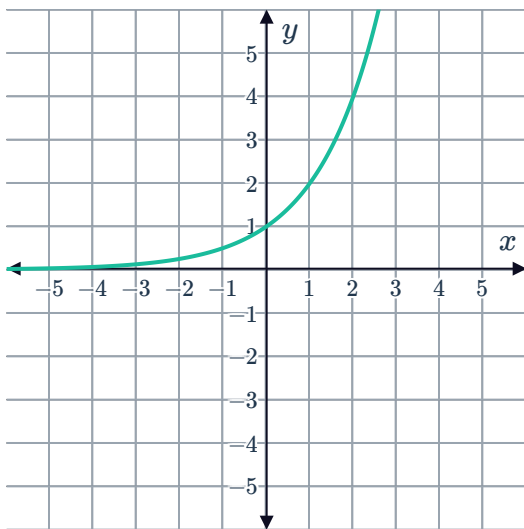
a



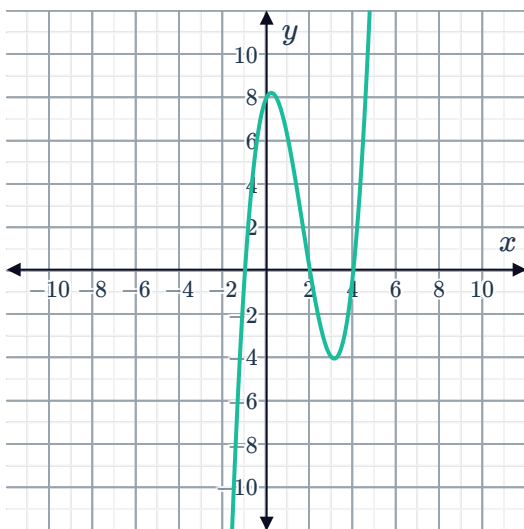
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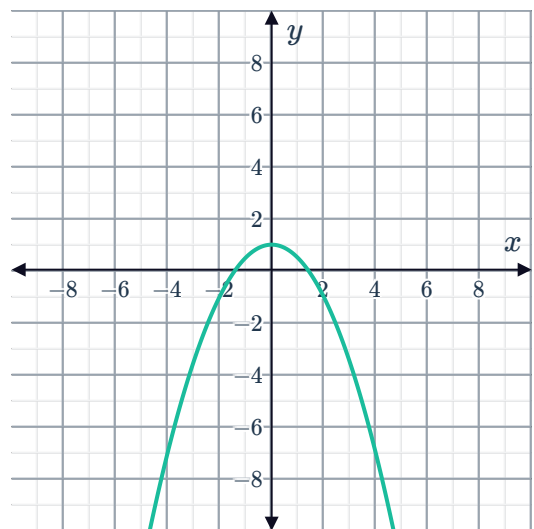
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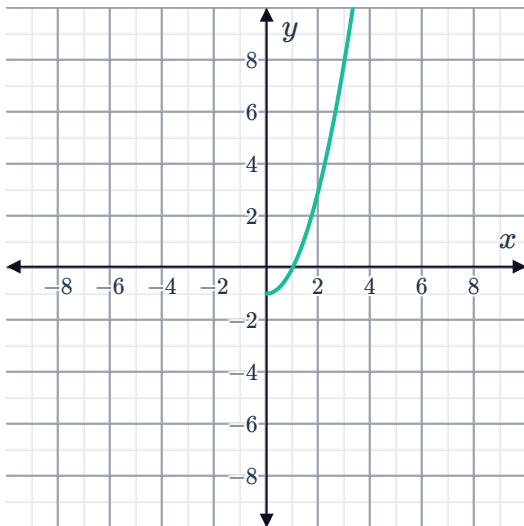
e



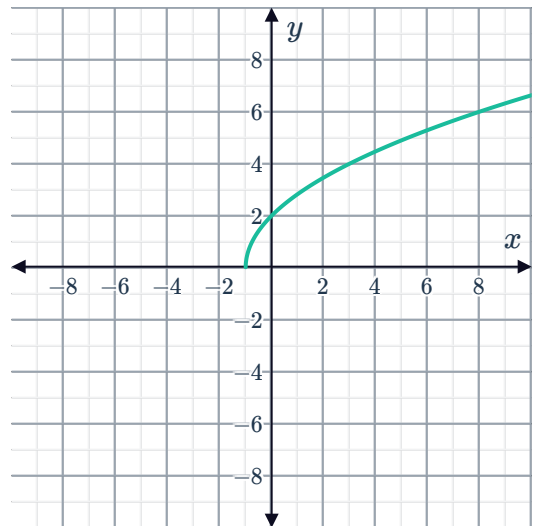
f



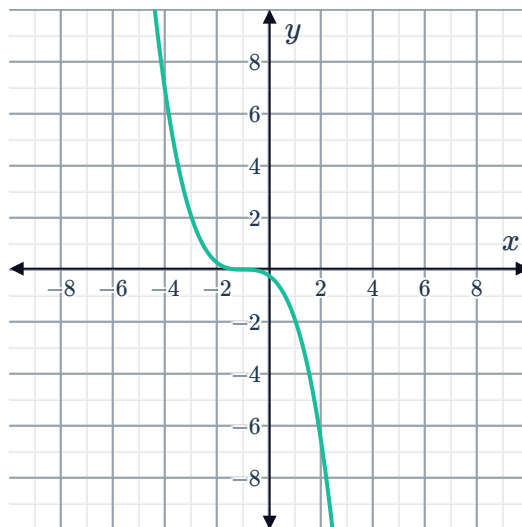
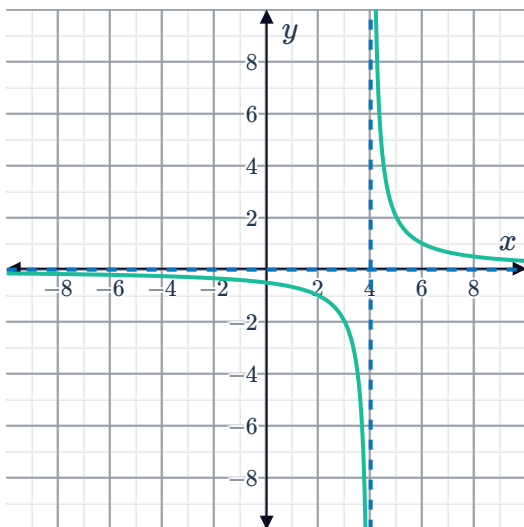
g



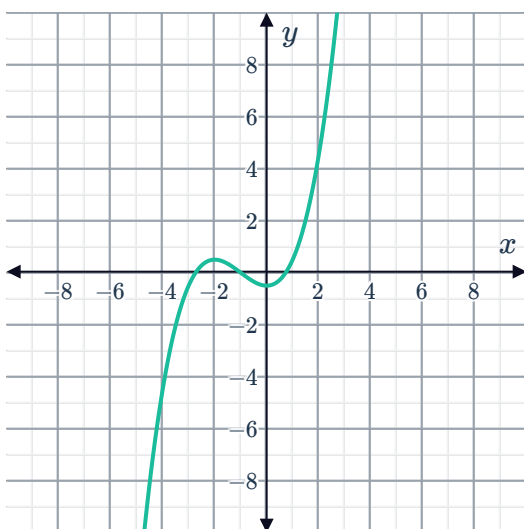
h



i

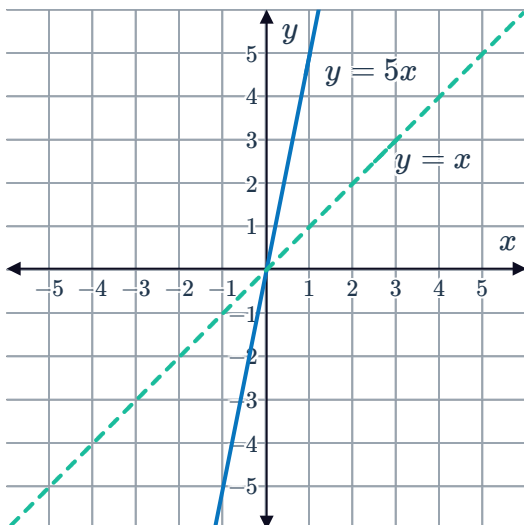


k

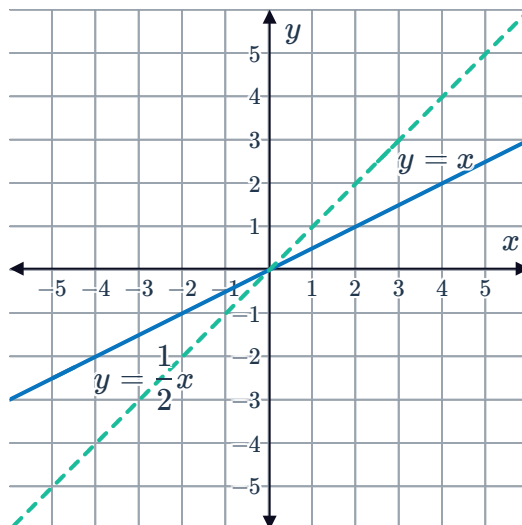


4 Sketch the graph of the inverse of the following functions:

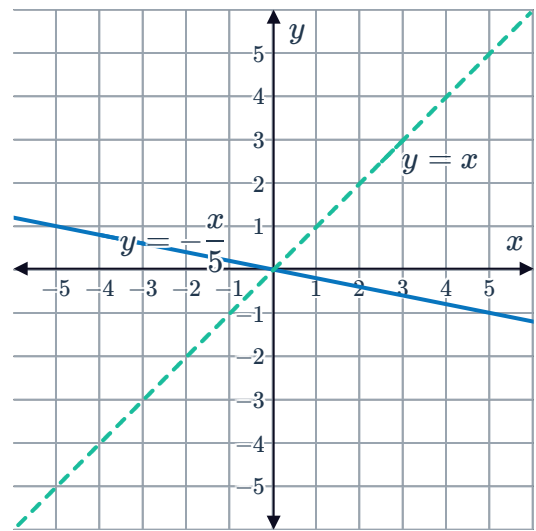
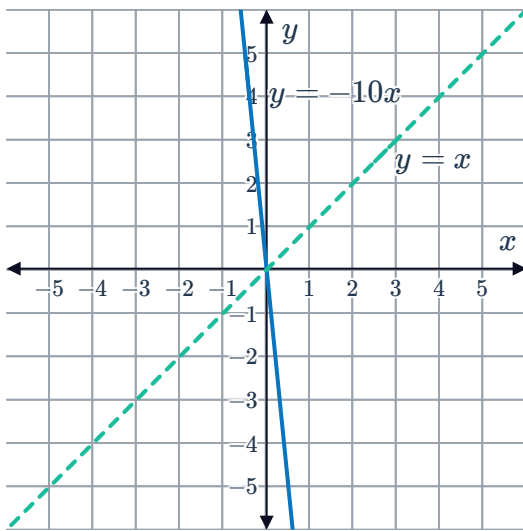
a



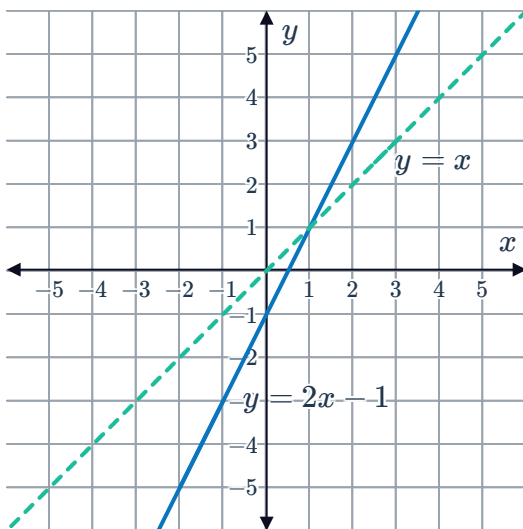
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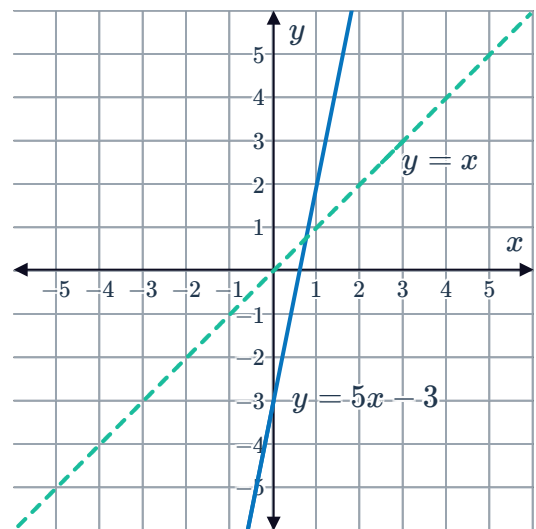
c



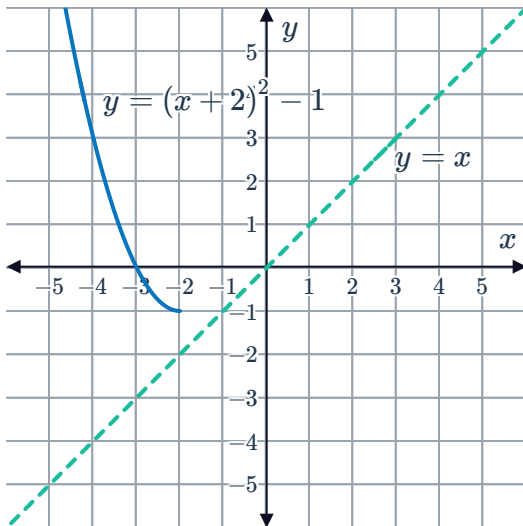
e



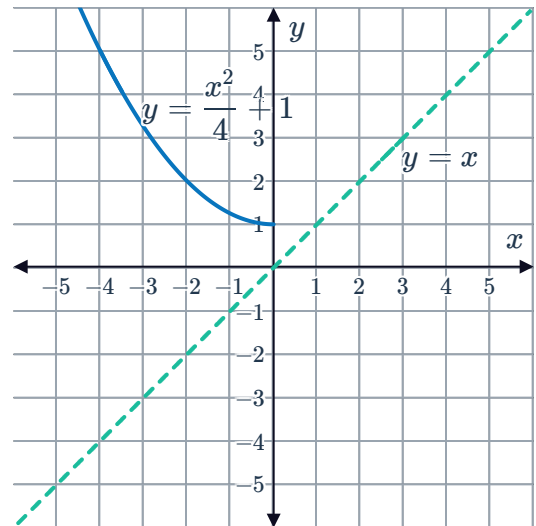
f



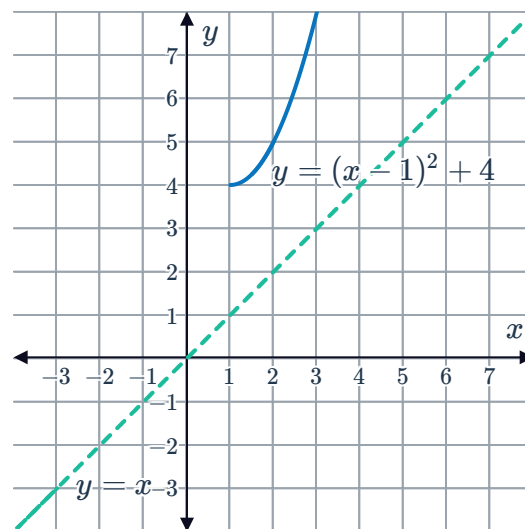
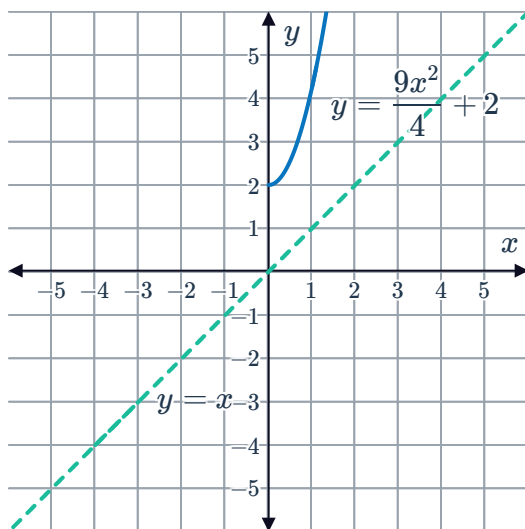
g



h

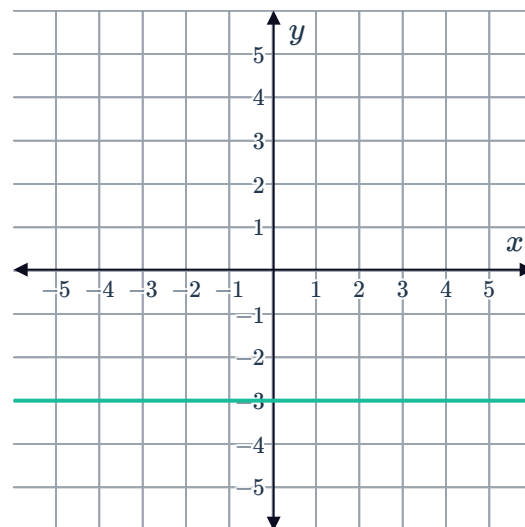


i



5 Consider the given graph:

- a Does this graph represent a function or just a relation?
- b If we try to invert this function, do we get a function or just a relation?



6 For each of the following pairs of functions:

i Find $f(g(x))$.

ii Find $g(f(x))$.

iii Are $f(x)$ and $g(x)$ inverses?

a $f(x) = -\frac{7}{2}x$ and $g(x) = -\frac{2}{7}x$

b $f(x) = 3x - 6$ and $g(x) = \frac{1}{3}x - 2$

c $f(x) = 3x + 12$ and $g(x) = \frac{1}{3}x - 4$

7 For each of the following functions, find an expression for the inverse function $y = f^{-1}(x)$:



a $f(x) = 5x$

b $f(x) = 7x - 8$

c $f(x) = \frac{4x}{3} - 5$

d $f(x) = 6x^3$

e $f(x) = 1 - \frac{x}{8}$

f $f(x) = \frac{x-7}{6}$

g $f(x) = \sqrt[3]{x-2}$

h $f(x) = (x-4)^3 + 3$

i $f(x) = \frac{5x}{x+2}$

j $f(x) = \frac{8}{x-9} + 5$

k $f(x) = \frac{8}{5x+3}$

l $f(x) = \frac{7}{(x+8)^3} - 3$

m $f(x) = 4(9x-2)^3 - 7$

Domain and range

8 The graph of a function f is defined by the single point $(-3, 6)$.

- a State the domain of the function f .
- b State the range of the function f .
- c The point is reflected about the line $y = x$ to form the graph of the function f^{-1} . State the coordinates of the single point that defines f^{-1} .
- d State the domain of the function f^{-1} .
- e State the range of the function f^{-1} .
- f What do you notice about the domain and range of f and f^{-1} ?

9 A one-to-one function f has a domain of $\{-7, 2\}$ and a range of $\{6, -5\}$. Find the:

- a Domain of f^{-1} .
- b Range of f^{-1} .

10 Consider the function f given by the set of ordered pairs $\{(5, 9), (7, -4), (8, 2)\}$.

- a State the domain of f .
- b State the range of f .
- c Find the inverse f^{-1} .
- d State the domain of f^{-1} .
- e State the range of f^{-1} .

11 Consider the function $f(x) = ax + b$ defined on $[-5, 5]$ where a and b are non-zero constants.

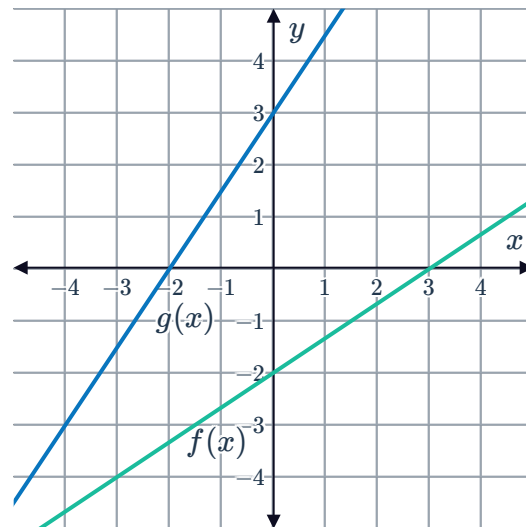
- a Using the fact that $f(1) = 2$ and $f^{-1}(1) = 3$, find the values of a and b .
- b State the domain of f^{-1} in interval notation.
- c State the range of f^{-1} in interval notation.

12 For each of the following functions, find:

- i $f^{-1}(x)$
 - ii The domain of $f^{-1}(x)$.
 - iii The range of $f^{-1}(x)$.
- a $f(x) = 5x + 15$ defined over all real values of x
 - b $f(x) = 4 - 3x$ defined over all real values of x
 - c $f(x) = 8x - 9$ defined over $[-4, 2]$
 - d $f(x) = x^2$ defined over $[0, \infty)$
 - e $f(x) = \sqrt{16 - x^2}$ defined over $[0, 4]$

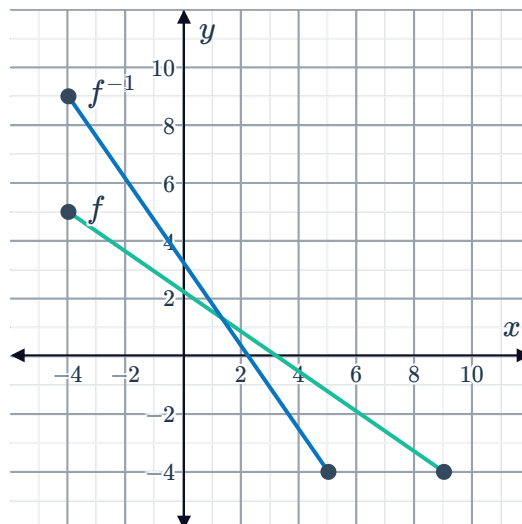
13 Consider the graphs of $f(x)$ and $g(x)$ shown:

- a State the equation of:
 - i $f(x)$
 - ii $g(x)$
- b Evaluate:
 - i $f(g(x))$
 - ii $g(f(x))$
- c State whether the following statements are true:
 - i $g(x)$ is an inverse of $f(x)$.
 - ii $f(x)$ is an inverse of $g(x)$.
 - iii $f(g(x))$ has slope $\frac{2}{3}$.
 - iv $g(f(x))$ has slope 1.



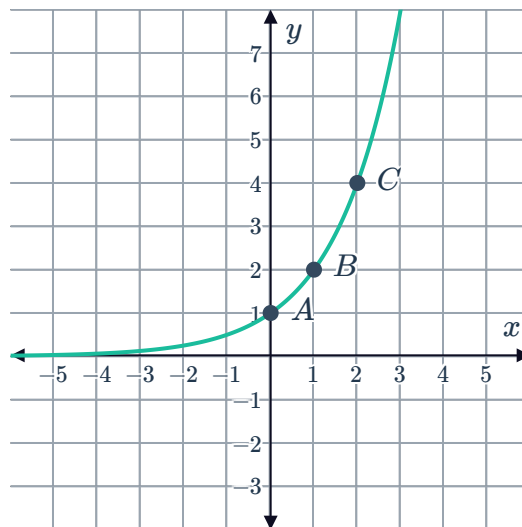
- 14 The graph of f is represented by the line segment joining $(-4, 5)$ and $(9, -4)$ while f^{-1} is represented by the line segment joining $(5, -4)$ and $(-4, 9)$.

- State the domain of f .
- State the range of f .
- State the domain of f^{-1} .
- State the range of f^{-1} .
- True or False: The range of f is the same as the range of f^{-1} .



- 15 Consider the graph of the function $f(x)$:

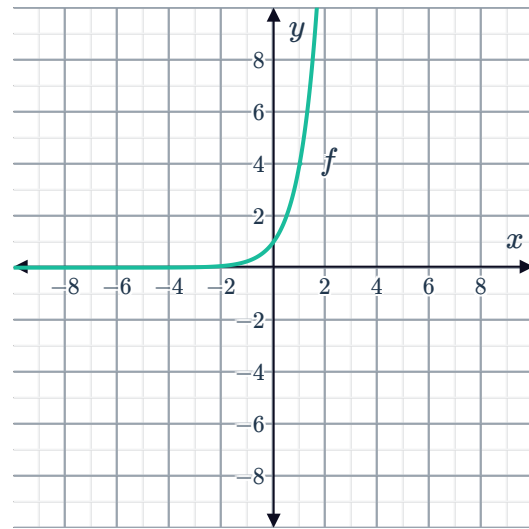
- State the coordinates of the points labeled:
 - A
 - B
 - C
- State the corresponding ordered pairs for f^{-1} , the inverse of f :
 - A'
 - B'
 - C'
- Plot the three inverse points on a coordinate plane.



16 Consider the graph of $f(x) = 4^x$:

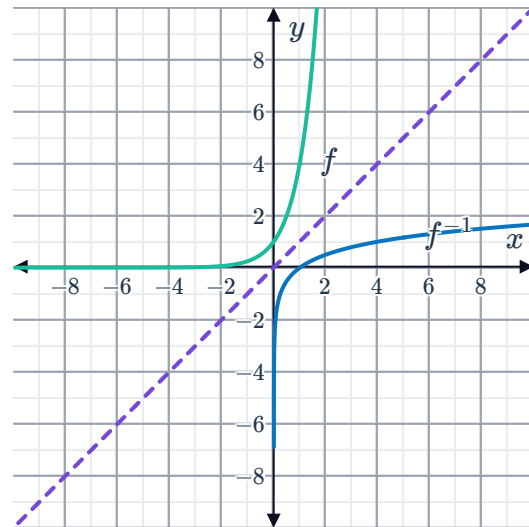


- a State the domain of the function f .
- b State the range of the function f .



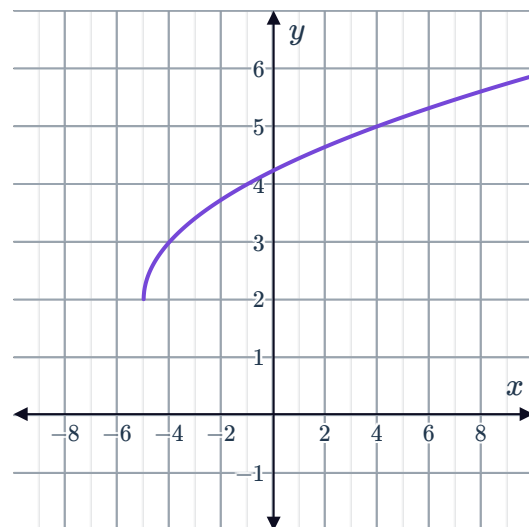
The graph of f is reflected about the line $y = x$ to produce the graph of f^{-1} as shown:

- c State the domain of the function f^{-1} .
- d State the range of the function f^{-1} .
- e What is the domain of f equivalent to?
- f What is the range of f equivalent to?



17 Consider the graph of $f(x) = \sqrt{x+5} + 2$:

- a State the domain of its inverse function.
- b State the range of its inverse function.
- c Find the inverse function.



18 Consider the function $f(x) = 0.8x + 0.8$.



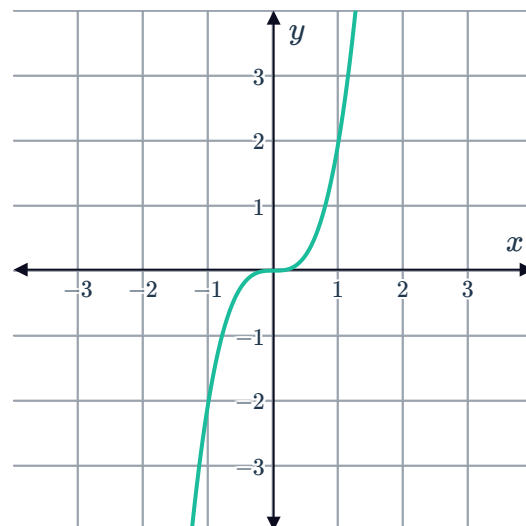
- a Sketch a graph of $f(x)$ by determining any two points on the line.
- b Sketch a graph of the inverse f^{-1} .
- c State the domain of f .
- d State the range of f .
- e State the domain of f^{-1} .
- f State the range of f^{-1} .

19 For each of the following functions:

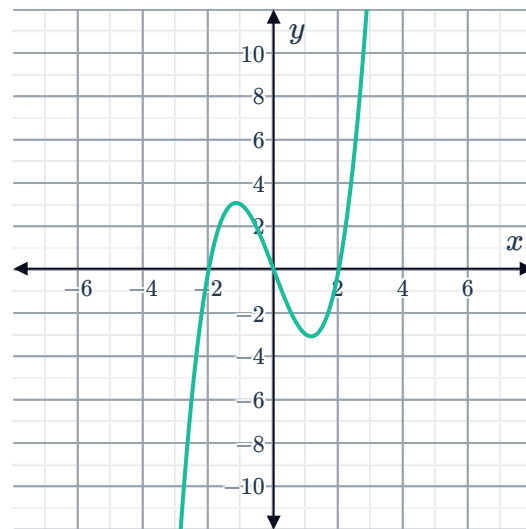
- i Sketch the function $f(x)$ over its domain.
 - ii Find the inverse, f^{-1} .
 - iii State the domain of f^{-1} .
 - iii State the range of f^{-1} .
 - iv Sketch the function f^{-1} over its domain.
- a $f(x) = x + 6$ defined over the interval $[0, \infty)$.
 - b $f(x) = 7 - x$ defined over the interval $[2, 9]$.
 - c $f(x) = (x - 6)^2 - 2$ defined over the interval $[6, \infty)$.
 - d $f(x) = \sqrt{4 - x}$ defined over the interval $[0, 4)$.
 - e $f(x) = (x + 2)^2 + 3$ defined over the interval $[0, \infty)$.

20 Consider the given graph:

- a Does this graph have an inverse function?
- b State the domain of this graph.
- c State the range of the inverse function of this graph.
- d Sketch the graph of the inverse function of this graph.



21 Consider the given graph:



- a** Does this graph have an inverse function?
- b** State the domain of this graph.
- c** State the domain restrictions that would allow the graph to have an inverse function.
- d** If the domain for the original function is restricted to only one of the restrictions, what would be the range of the inverse function?

22 The tax on a new tablet is 7% of the advertized price, A .

- a** Determine the equation for the total cost T as a function of the advertized price A .
- b** Now, express the advertized price A as a function of the total cost T .